

Cybersecurity

Asymmetric Encryption and Hashing

Lesson 11.2



Learning Objectives

By the end of today's lesson, you will be able to:

1

Calculate the required number of symmetric and asymmetric keys based on the number of people exchanging secure messages.

2

Use GPG to generate keys and encrypt and decrypt private messages.

3

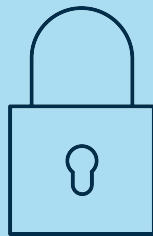
Use hashes to validate the integrity of data.

4

Use digital signatures to validate the authenticity of data.



Welcome



Cryptography Review

Cryptography Review

01

Cryptography

02

Ciphers

03

Encryption/Decryption

04

Key

05

Block and stream ciphers

06

Encoding

07

PAIN

08

Security trade-off

09

Symmetric key algorithms

10

OpenSSL



Questions?





Activity

Cryptography refresher

You will continue to act as a security analyst working for the Hill Valley Police Department.

In this review activity, you must encrypt and decrypt a message using **OpenSSL**.

Suggested Time
20 minutes

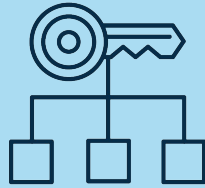


Questions?





Time's Up!
Let's review



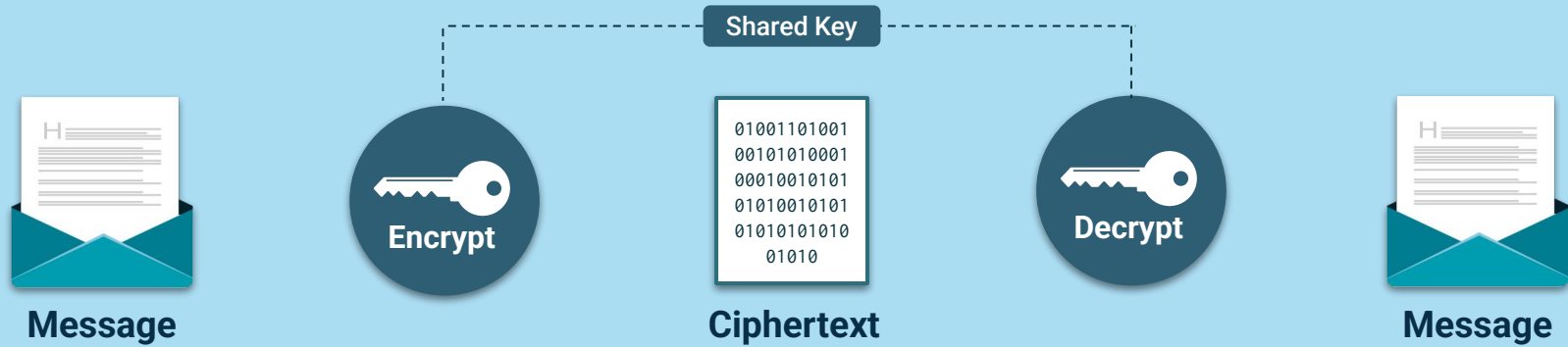
Key Management and Exchange

Key Management and Exchange

In the previous activity, we used **symmetric key encryption**.

While its benefits include speed, efficiency, and simplicity of use, symmetric encryption also has disadvantages.

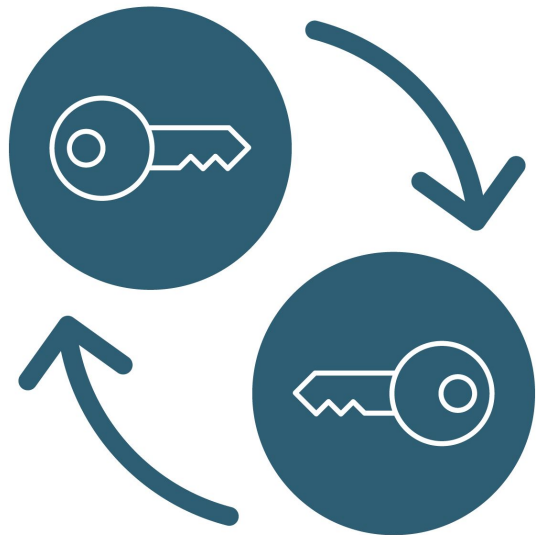
Symmetric Encryption





Key Management and Exchange

Disadvantage 1: Secure key exchange



Offline Exchange (Out-of-Band Exchange)

Can include mailing the key, or calling the other party and reading the key.

- Intercepted mail
- Tapped phone line

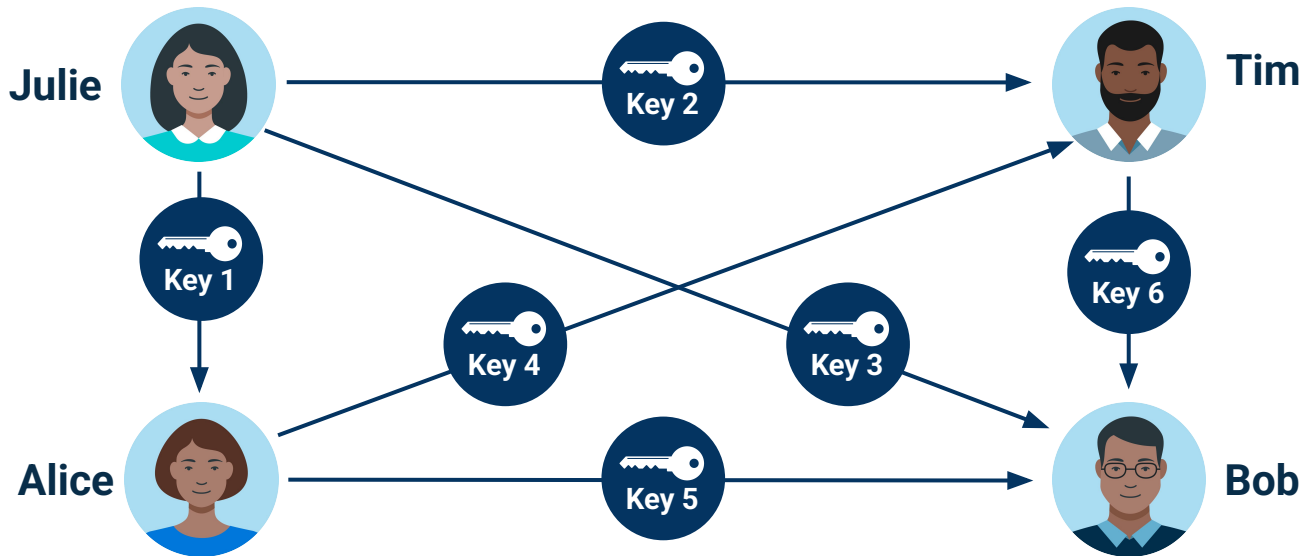
Diffie-Hellman Key Exchange

Uses mathematics to create a shared secret between two parties over a public channel.



Key Management and Exchange

Disadvantage 2: Key management



Symmetric key cryptography between two parties only requires managing one key. But there needs to be a key **for each combination of individuals**.



Key Management and Exchange

Count of symmetric keys = $(N * (N-1)) / 2$

For an organization with 7 employees:

$$(7 * 6) / 2 = 42 / 2 = 21$$



For an organization with 1,000 employees:

$$(1,000 * 999) / 2 = 499,500$$





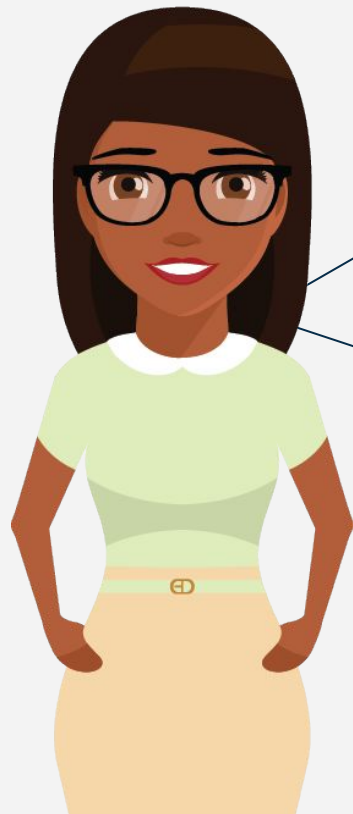
Asymmetric/Public Key Cryptography



Symmetric key disadvantages
can be addressed with
asymmetric key encryption.

Asymmetric Key Encryption

Each individual possesses a two-key pair:



Private keys are kept secret and can affect the confidentiality of messages if exposed.

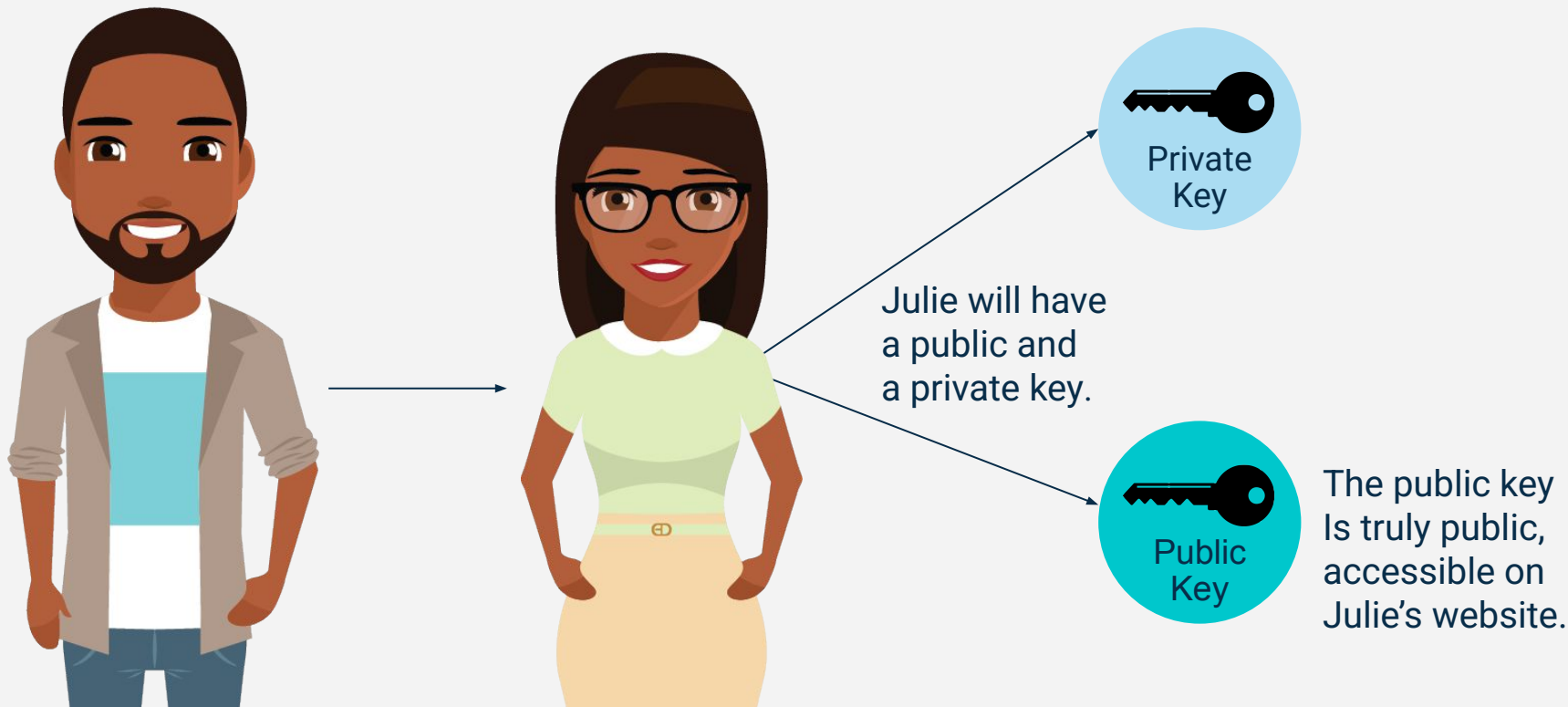


Public keys are public and accessible to all.

Like symmetric keys, these public and private keys are typically a random alphanumeric string.

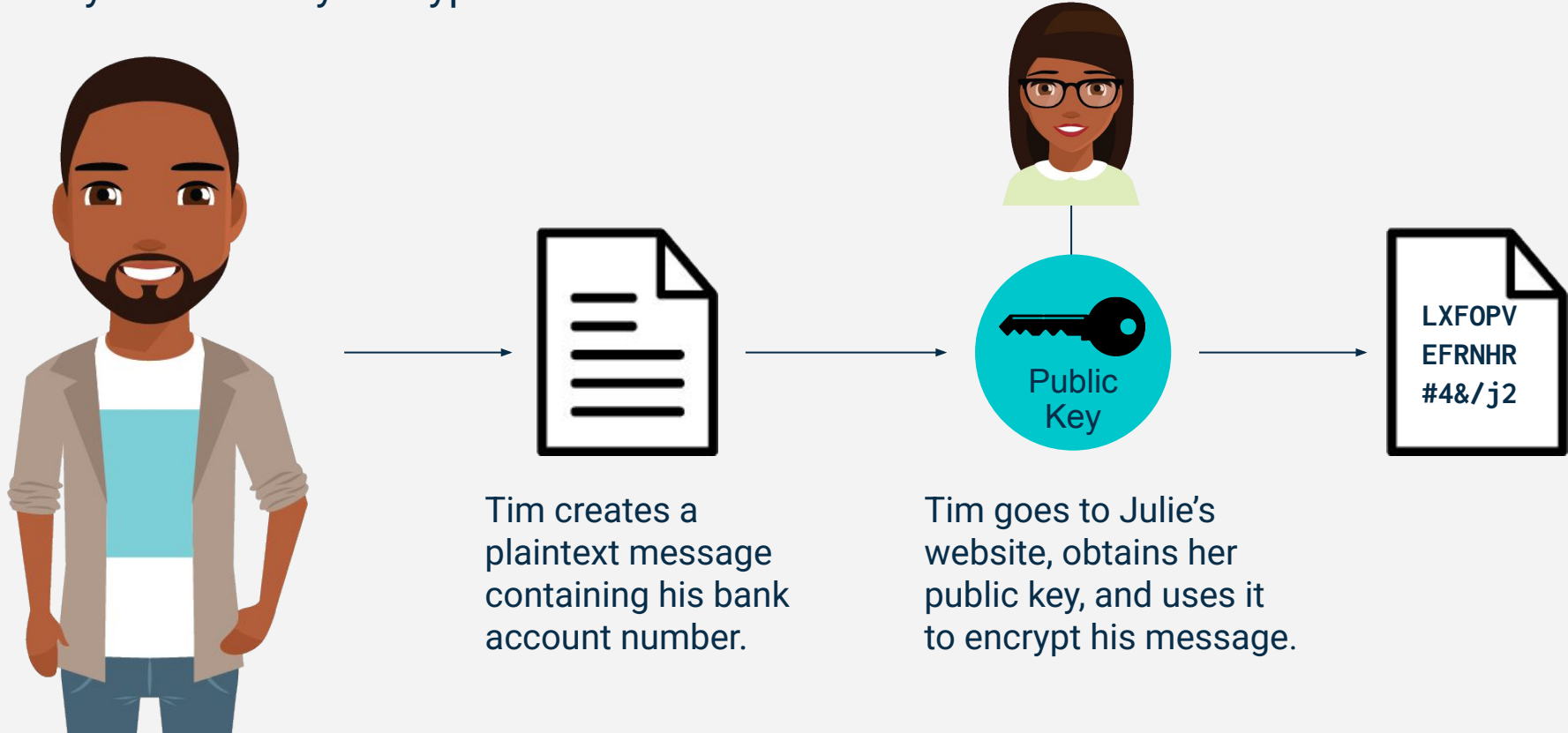
Asymmetric Key Encryption

Scenario: Tim wants to send Julie his bank account number using asymmetric key encryption.



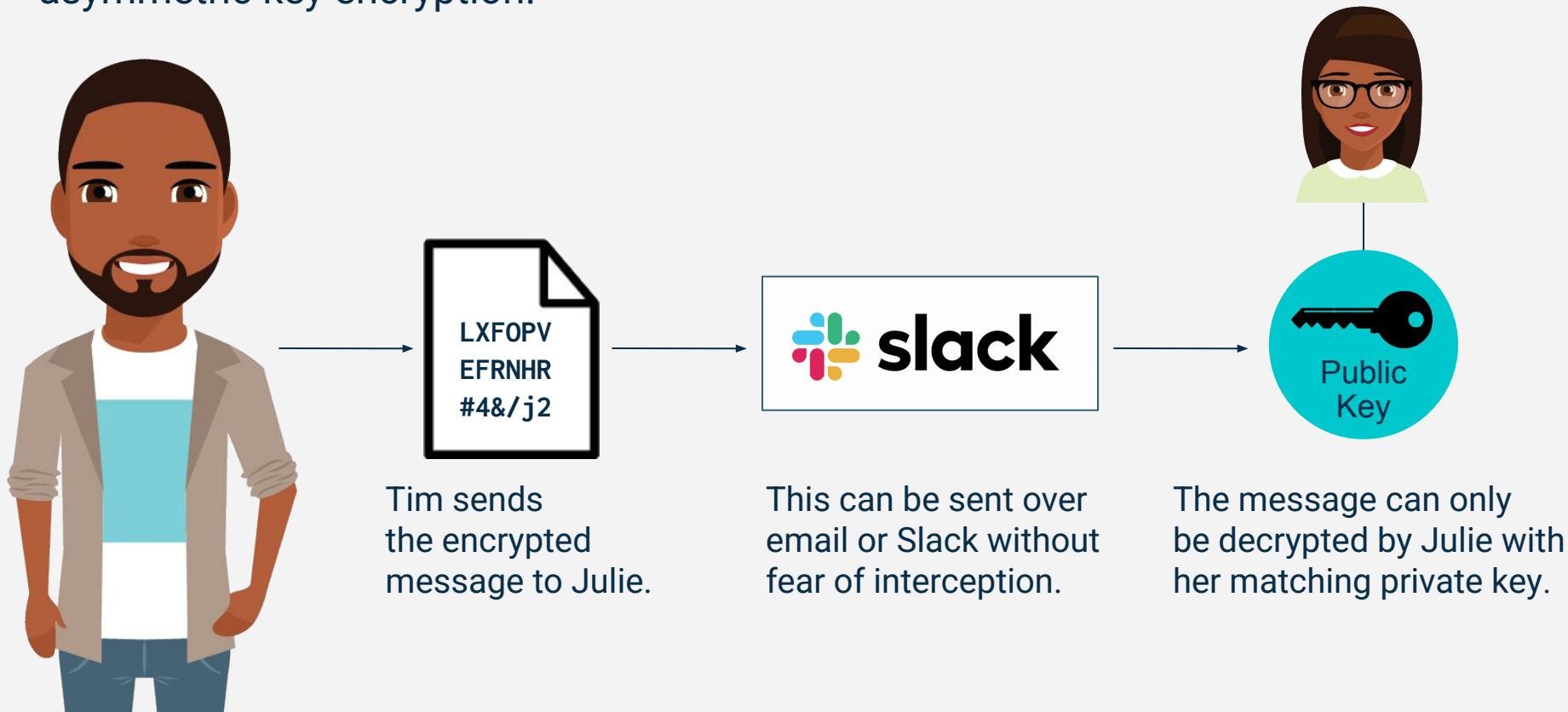
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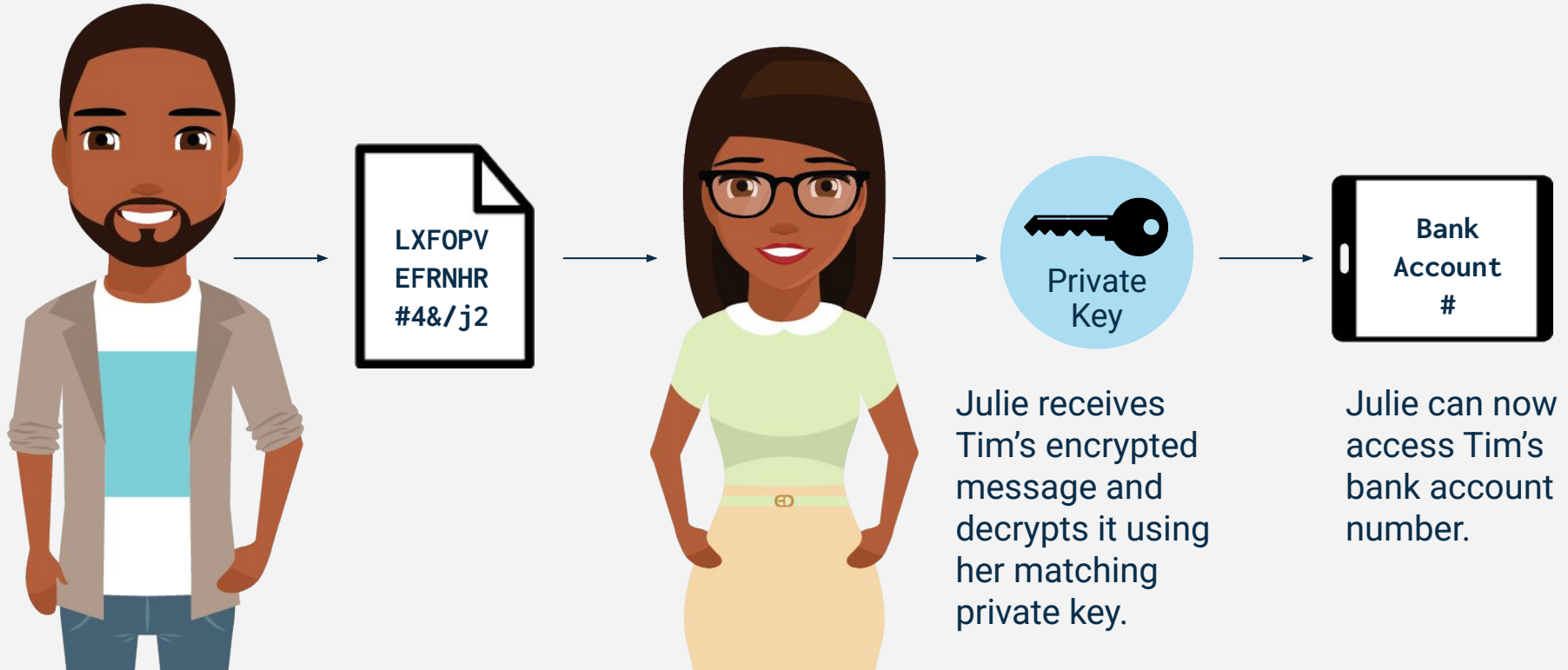
Asymmetric Key Encryption

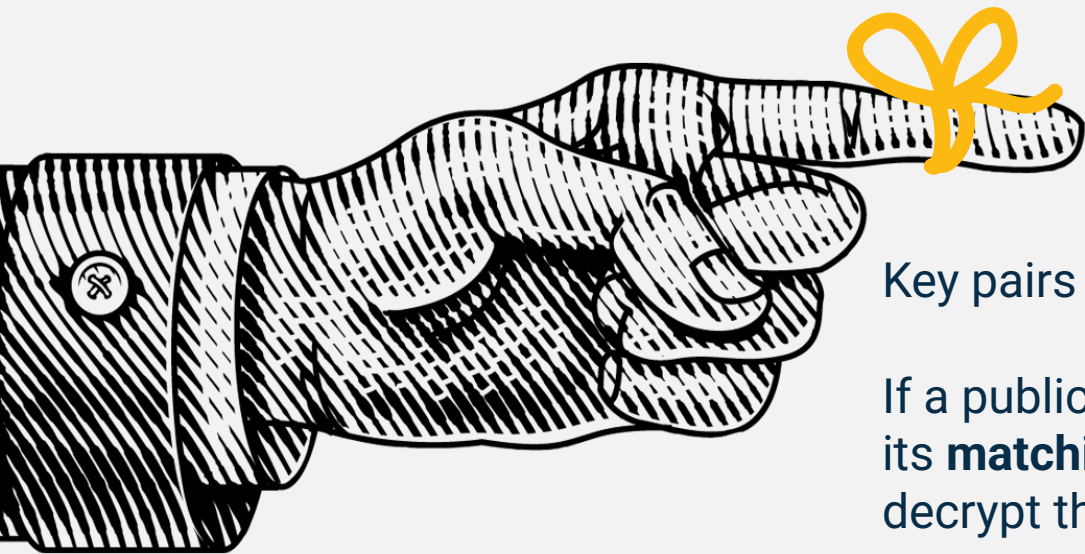
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Asymmetric Key Encryption

Scenario: Tim wants to send Julie his bank account number using asymmetric key encryption.





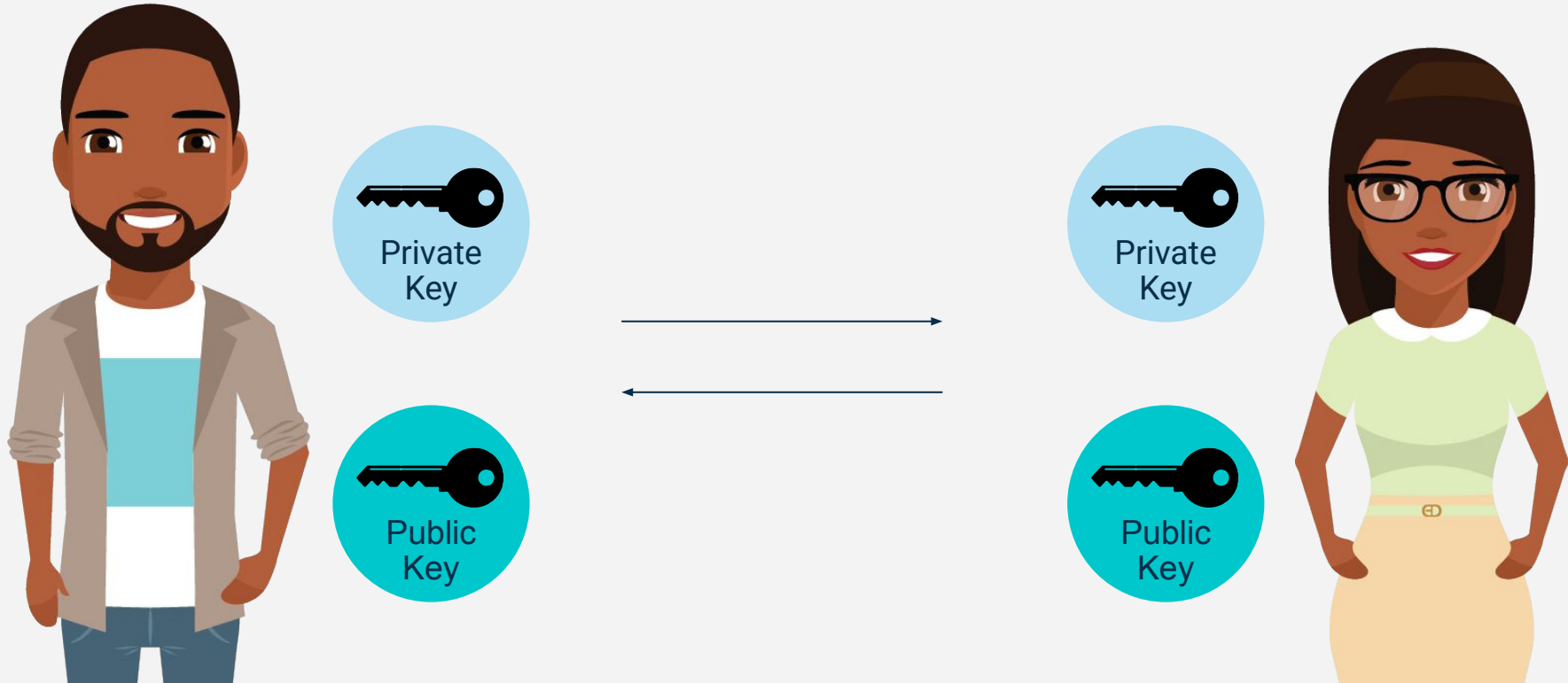
Remember

Key pairs **always** have to be used together.

If a public key is used to encrypt a message, its **matching private key** will be used to decrypt the message.

Asymmetric Key Encryption

If Tim and Julie each needed to send messages to each other, **four keys** would be required for secure communication.



Key Management

While symmetric key count grows exponentially larger as the number of employees increases, the asymmetric count stays proportionate and manageable.

For a company with 12 employees:



Symmetric

$$\begin{aligned} & (N * (N-1)) / 2 \\ & (12 * 11) / 2 = 66 \end{aligned}$$

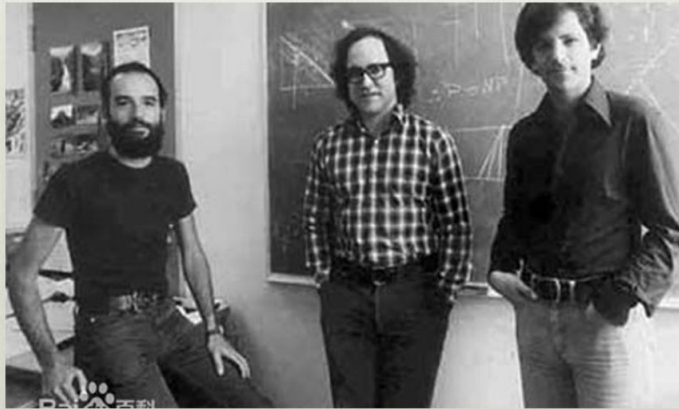
Asymmetric

$$\begin{aligned} & N * 2 \\ & (12 * 2) = 24 \end{aligned}$$

RSA

Asymmetric encryption uses an algorithm called RSA, which was introduced in 1977 and named after the surnames of its creators.

Ron Rivest, Adi Shamir and
Leonard Adleman

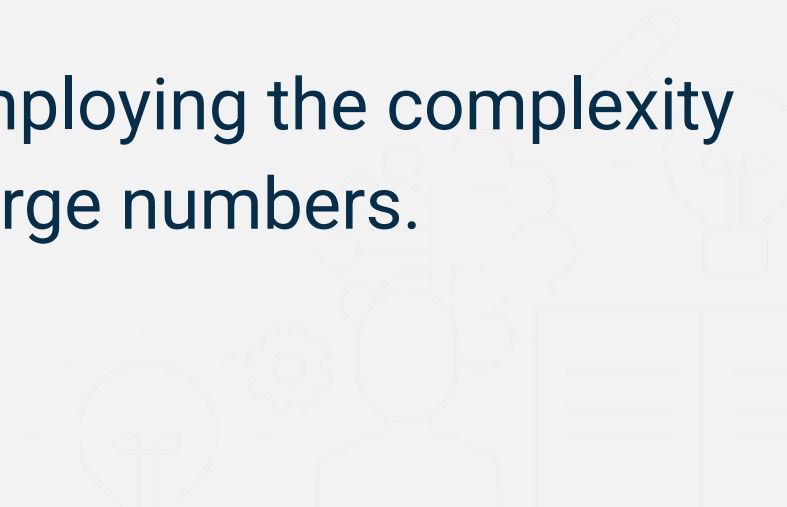


Popularly known as **Rivest-Shamir-Adleman (RSA)** algorithm, it was built after **Whitefied Diffie, Martin Helman and Ralph Merkle** introduced public key cryptography in **1976**. RSA is an **asymmetric cryptography** popularly known as **public key cryptography**.



RSA is an asymmetric algorithm that is still a worldwide standard today.

It works by employing the complexity of factoring large numbers.



Asymmetric Key Summary

Asymmetric encryption uses **two keys**: One public and one private

- Public keys are accessible by anyone
- Private keys are only accessible by their owner
- Public and private keys have to work together
- Asymmetric encryption addresses the challenges of symmetric encryption
- The standard asymmetric algorithm used today is RSA



Questions?





Applying Public Key Cryptography with **GPG**

GNU Privacy Guard (GPG)

GPG is:

Command Line Tool

- Create keys
- Encryption
- Decryption

Free Program

- Multiple distros
- Symmetric algorithms
- Asymmetric algorithms





Demonstration Scenario

We will demonstrate how to create a key pair and do asymmetric encryption and decryption with GPG.

Asymmetric Encryption

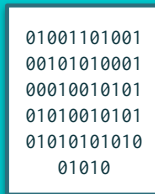


Message

Public Key



Encrypt



Ciphertext

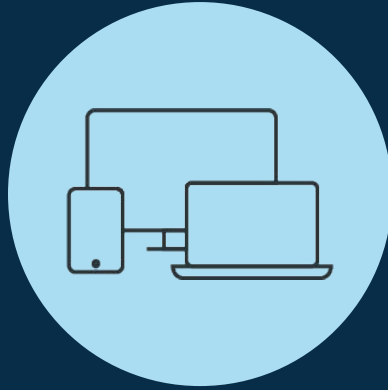
Private Key



Decrypt



Message



Instructor Demonstration

GPG



Activity

GPG

You will continue to play the role of a security analyst working for the Hill Valley Police Department.

- Captain Strickland has implemented your suggestion of using asymmetric key encryption across all departments at Hill Valley.
- You will use asymmetric encryption with GPG to encrypt and decrypt text files.

Suggested Time
20 minutes



Time's Up!
Let's review

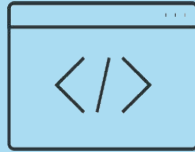


Questions?





Break
15 mins



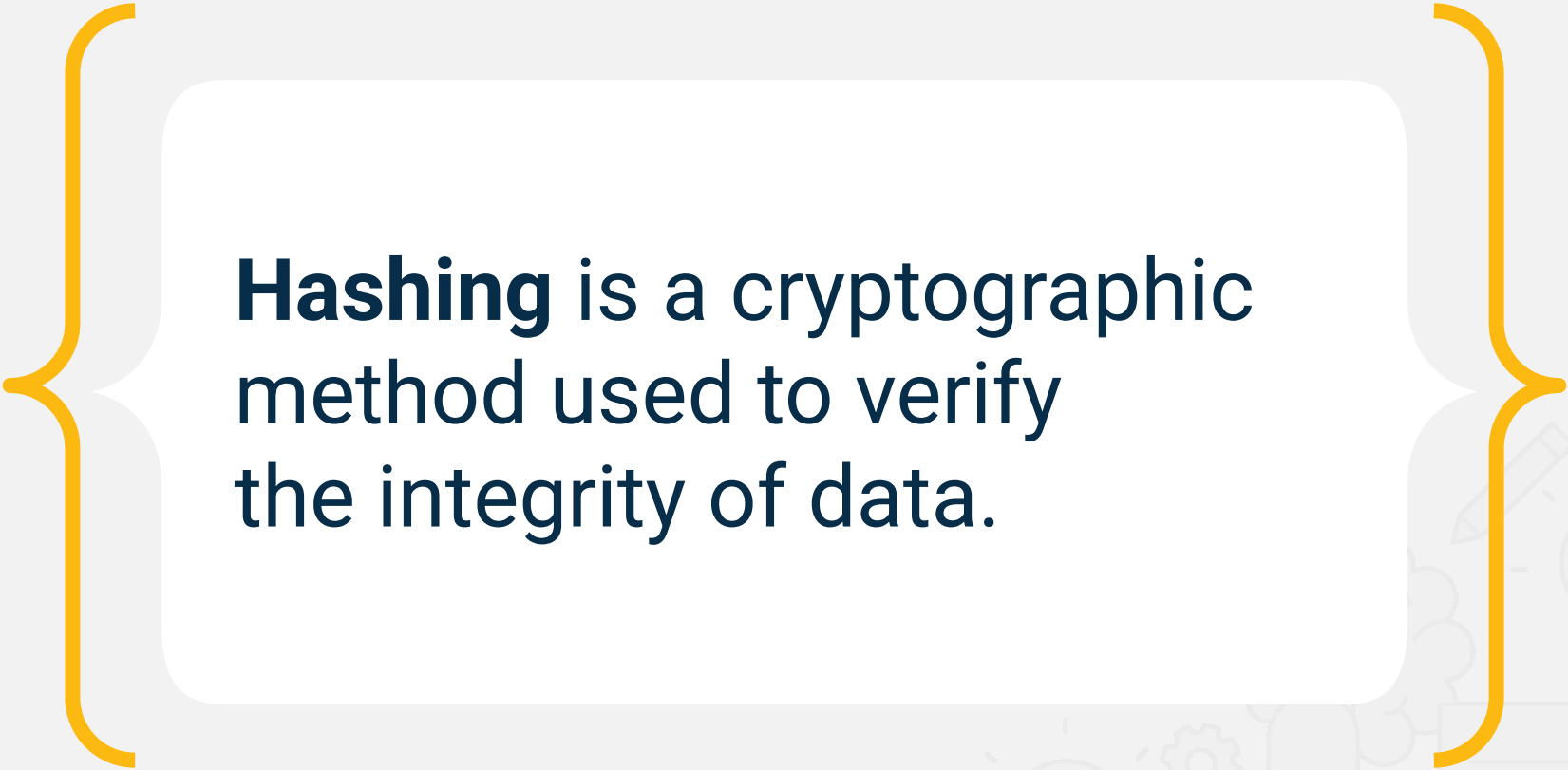
Hashing and Data Integrity

Always Remember the CIA!

So far, we've used cryptography to protect privacy and confidentiality.

It can also be applied to protect the **integrity** of data.





Hashing is a cryptographic method used to verify the integrity of data.



Hashing

Similar to encryption, **hashing** uses mathematical algorithms called **hashing algorithms** to take data and generate a unique output.

I Love Cryptography!

MD5



676e4bfff90a76853bda00773f7ad4bed

Difference 1: Keys vs. No Keys

Encryption takes plaintext and converts it to ciphertext with a key and an algorithm.



Plaintext



Encryption Key



Encrypted Text



Decryption Key

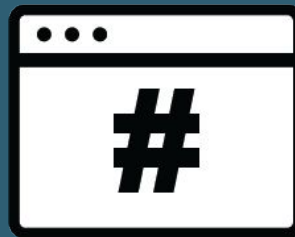


Plaintext

Hashing takes plaintext and converts it to a message digest with an algorithm and no key.



Plaintext



Hash Function



Hashed Text

Difference 1: Keys vs. No Keys

Message digests are also known as fingerprints, hashes, and checksums.

They are unique identifiers of the plaintext that is outputted from the algorithm.

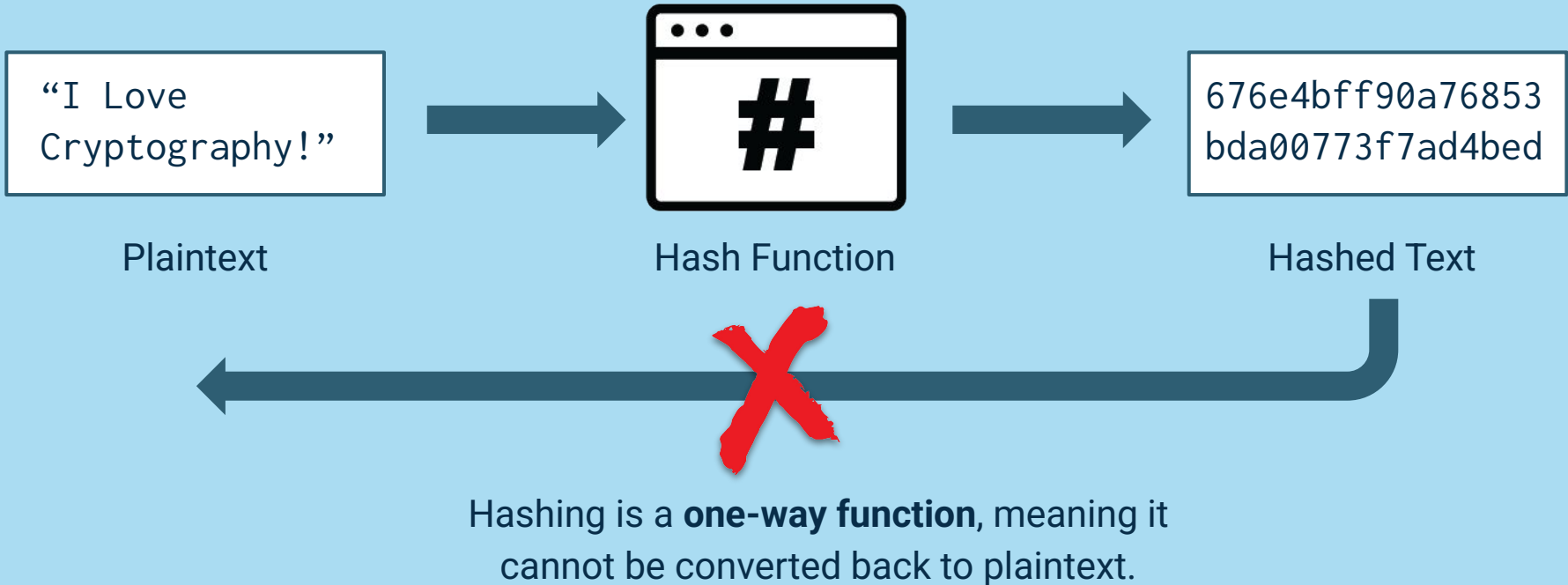
For example:

```
676e4bff90a76853bda00773f7ad4bed
```



Difference 2: Reversible vs. Irreversible

Encryption converts ciphertext and plaintext back and forth using encryption and decryption.



Difference 3: Output Lengths

Hashing algorithms output fixed lengths. Regardless of the input length, the output length is always the same.

I Love Cryptography!

#

676e4bff90a76853bda00773f7ad4bed

Hi

#

c1a5298f939e87e8f962a5edfc206918

We hold these truths to be self-evident, that all men are created equal, that they are endowed by their Creator with certain unalienable Rights, that among these are Life, Liberty, and the pursuit of Happiness...

#

124650d689d47f777f715a25260a29c2

This is the hash of the entire U.S. Declaration of Independence!

Difference 4: Goals



Encryption's
main goal is
privacy.

Hashing's
main goal is
integrity.

Difference 4: Goals

If a small change is made to the sentence, the same hashing algorithm would produce a completely different message digest.

I Love Cryptography!

#

676e4bffa90a76853bda00773f7ad4bed

I Love Cryptography!!

#

4e6fc433ff57a6c4a854cbbef65f61a

Hashing Algorithms

Hashing has several algorithms:

SHA	Stands for secure hash algorithms and includes its successors SHA1 and SHA2 .
SHA2	Has variations with different security strengths: SHA-256 and SHA-512 .
MD	Stands for message digest and has several variations: MD2 , MD4 , and MD5 .
LM and NTLM	Are hashes used by Windows.



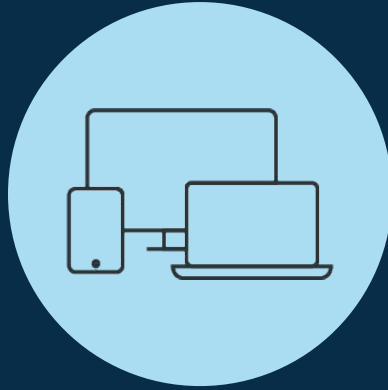
Demonstration Scenario

There are **command-line tools** that easily create message digests with a simple terminal command.

- **md5sum** uses the MD5 hashing algorithm
- **sha256sum** uses the SHA-256 hashing algorithm

```
sysadmin@ubuntu:~$ md5sum
```

```
sysadmin@ubuntu:~$ sha256sum
```



Instructor Demonstration

Creating hashes on the command line

Hashing and Data Integrity Summary



While encryption is used for confidentiality, hashing is used for integrity.



Hashing uses hashing algorithms to create message digests from plaintext.



A small change in the plaintext causes a significant change in the message digest.



Common hashing algorithms are SHA1, SHA2, and MD5.



`md5sum` and `sha256sum` are command-line tools used to create message digests and check the integrity of files.



Activity

Generating hashes

In this activity, you will continue to play the role of a security analyst working for the Hill Valley Police department.

- Captain Strickland would like you to determine if any investigation report files have been changed, as well as what was changed.
- Your task is to generate hashes of each backup and current file, and compare the hashes to determine which files were changed.

Suggested Time
17 minutes

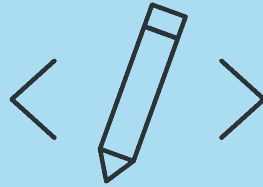


Time's Up!
Let's review



Questions?





Digital Signatures



Cryptography can also
be used to validate
authenticity.





Digital Signatures

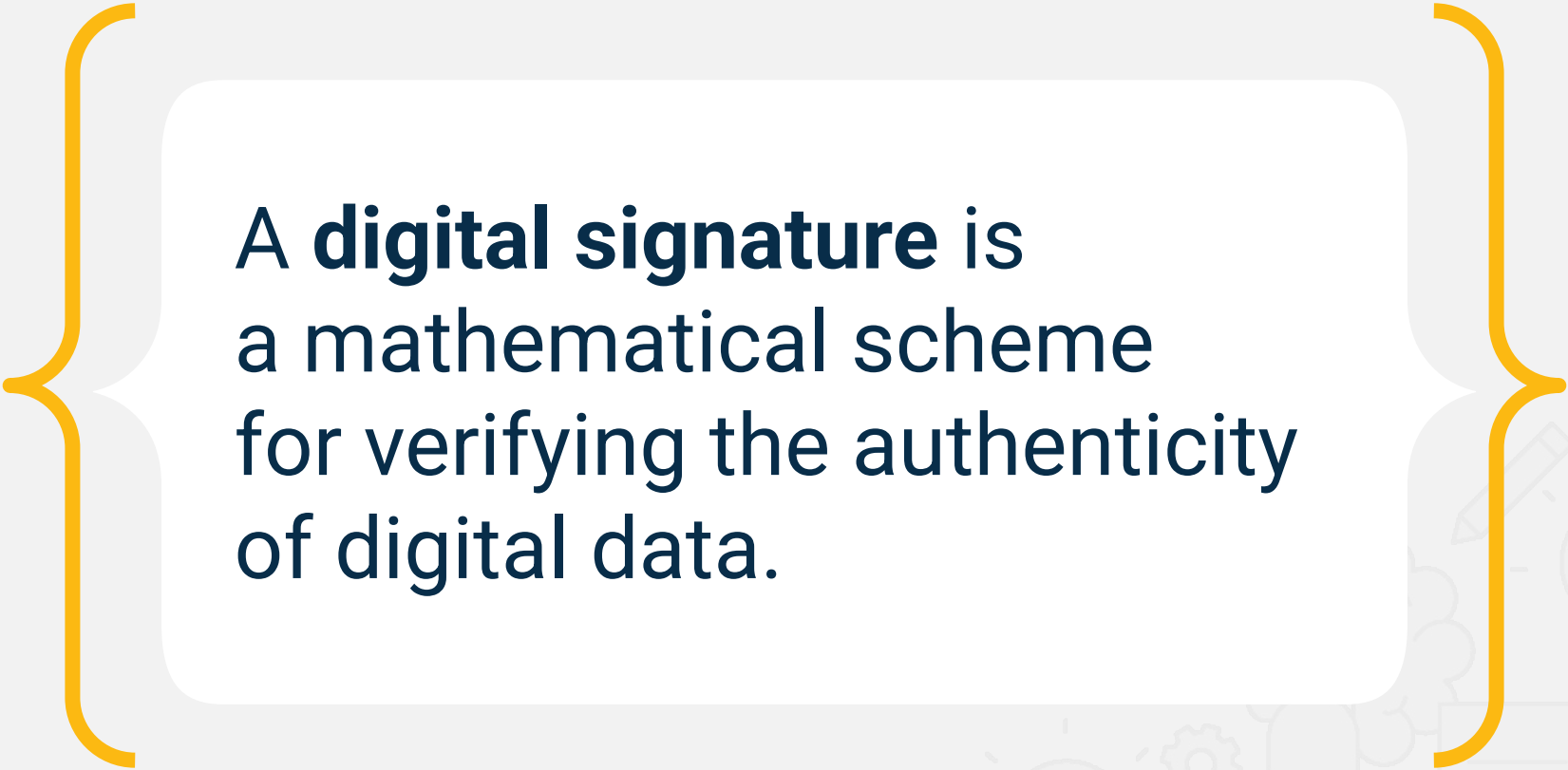
Authenticity

For example:

An accounting representative receives a message from their CEO to wire funds immediately to a location in Asia.

The accounting representative needs to validate that the message is **authentic** and actually from the CEO.





A digital signature is
a mathematical scheme
for verifying the authenticity
of digital data.





Digital Signatures

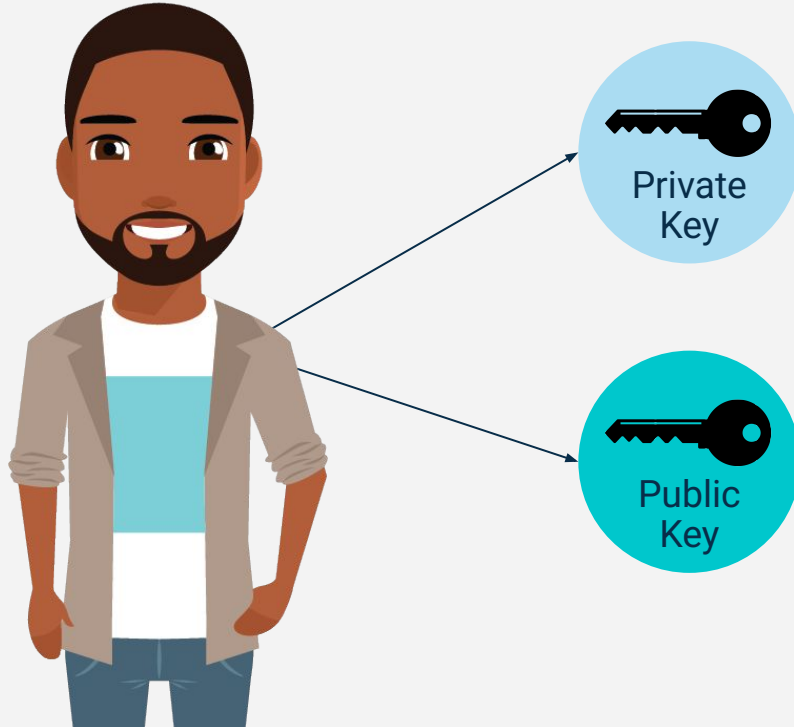
Authenticity, non-repudiation, and integrity

- Digital signatures are primarily used for authenticity, but can also provide:
 - Non-repudiation
 - Integrity
- In the U.S. and several other countries, digital signatures are considered **legally binding**.
- Similar to asymmetric encryption, digital signatures also use public key cryptography, but in **reverse**.



Digital Signatures

Scenario: Tim wants to send Julie a message that says “Transfer \$500 to the account I provided to you” and then digitally sign the message.



STEP 1

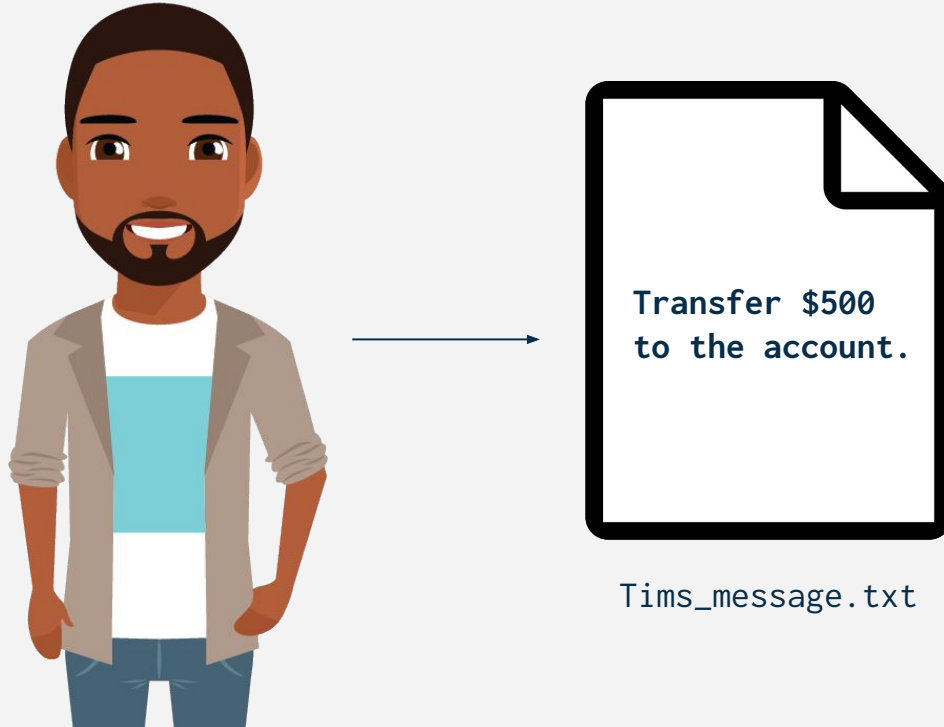
Key Creation

Tim will have a public and a private key.

The public key is truly public and accessible on his website.

Digital Signatures

Scenario: Tim wants to send Julie a message that says “Transfer \$500 to the account I provided to you” and then digitally sign the message.



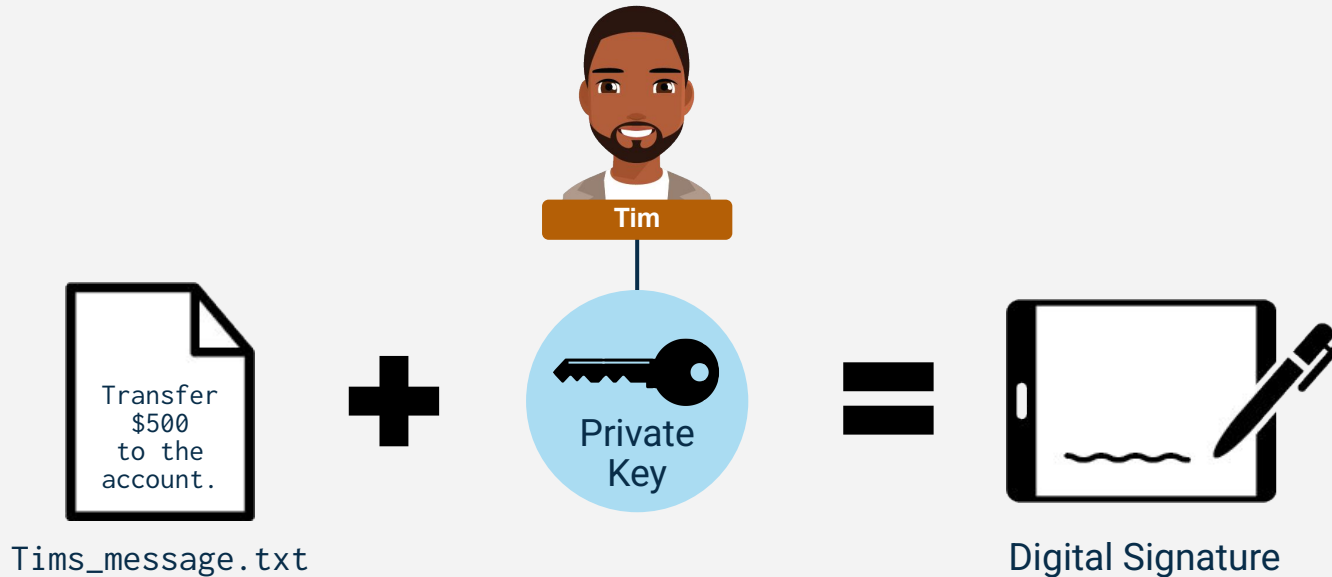
STEP 2

Creating the Message

Tim places his message inside a file.

Digital Signatures

Scenario: Tim wants to send Julie a message that says “Transfer \$500 to the account I provided to you” and then digitally sign the message.



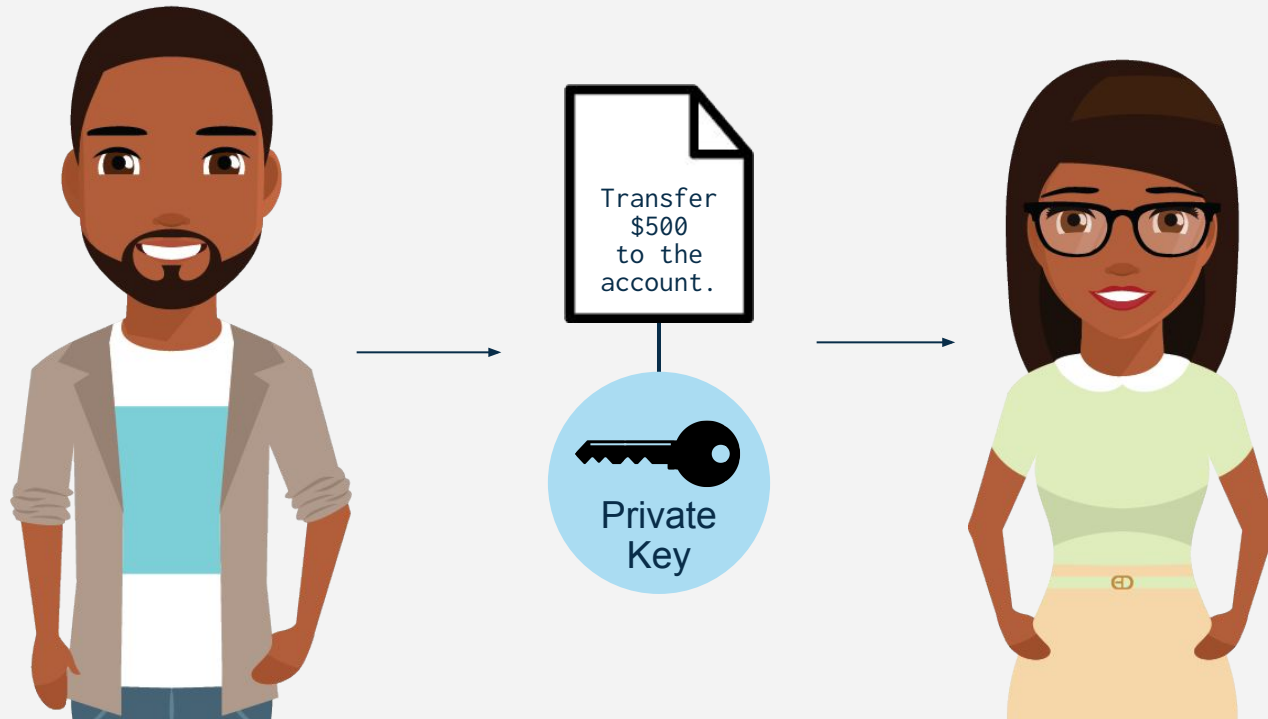
STEP 3

Signing the Message

Tim signs the message with his private key in order to create a digital signature.

Digital Signatures

Scenario: Tim wants to send Julie a message that says “Transfer \$500 to the account I provided to you” and then digitally sign the message.



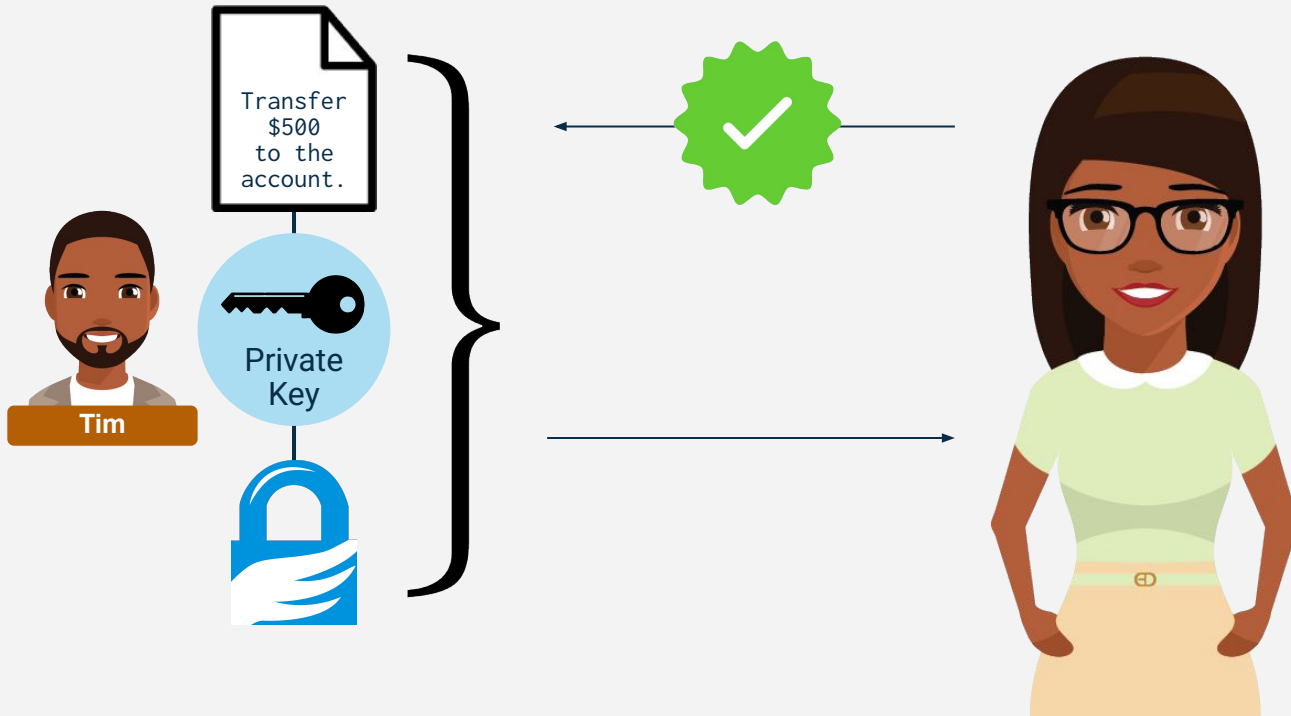
STEP 4

Sending the Message

Tim sends the digital signature to Julie along with his plaintext message.

Digital Signatures

Scenario: Tim wants to send Julie a message that says “Transfer \$500 to the account I provided to you” and then digitally sign the message.



STEP 5

Validating the Signature

Julie uses Tim's public key to validate the signature, using a validation tool such as GPG.



Tim and Julie Scenario

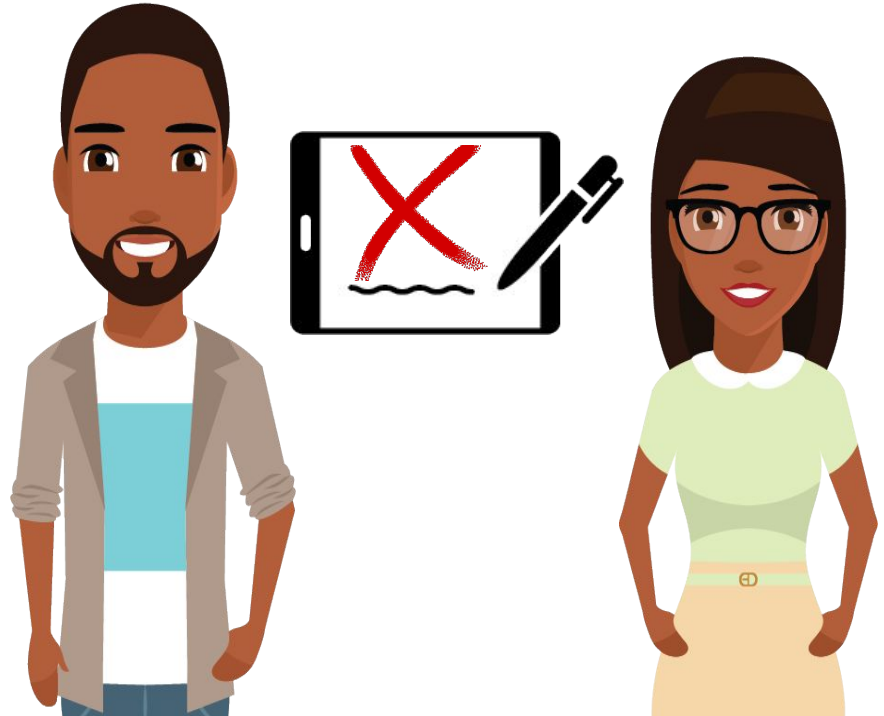
Integrity:

If Tim's message was modified, the digital signature validation would fail.

Non-repudiation:

Tim would not be able to deny he was the one who signed the message.

This would be an example of a **detached signature**.



Digital Signatures

Other digital signatures include:

All at Once

A signature appended to an encrypted message.

Clearsigned

A signature appended to an unencrypted message.

Signed Hash

Instead of signing a message, a hash is created first and the hash is signed for verification.



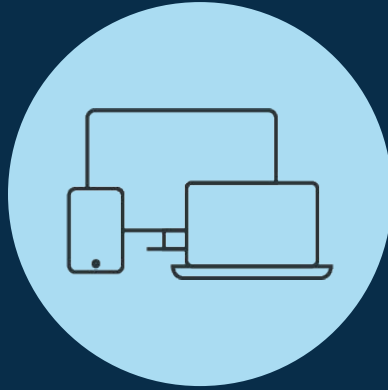
Demonstration Scenario



In the next demo, we will apply a detached digital signature with **GPG**.

This is the same GPG program used for GPG asymmetric encryption.

Let's take a look!



Instructor Demonstration

Signing with GPG



Activity

Digital signatures

In this activity, you will continue to play the role of a security analyst working for the Hill Valley Police department.

- Captain Strickland has sent you several more urgent messages about the Alphabet Bandit case.
- It's quite likely that one of the messages was sent by the Alphabet Bandit pretending to be Captain Strickland to throw off the investigation.
- Your task is to use GPG to determine which of the messages is **not authentic**.

Suggested Time
20 minutes



Time's Up!
Let's review



Questions?





Next Class

Applied Cryptography and Cryptographic Attacks



Thank You